Chapter 1

Advanced MCUs and Microprocessors (MPUs)

ENEL712 Embedded Systems Design — Lecture 4

Summary

Moving beyond 8-bit MCUs: hybrid 8/16-bit XMEGAs, hybrid DSP/MCUs, the ARM Cortex hierarchy, and how MCUs differ from MPUs in integration, performance, and complexity.

Topics Covered

- Transitioning beyond 8-bit MCUs
- ARM architecture and Cortex hierarchy
- MCUs vs MPUs
- Selection advice for advanced systems

1.1 Transitioning Beyond 8-bit MCUs

When projects require higher raw throughput (trig, multiplication), better timing, or complex comms, 8-bit MCUs may not be enough.

1.1.1 Atmel XMEGA MCUs

- Hybrid 8/16-bit: 8-bit data bus with 16-bit ALU; better integer performance for values >255.
- Event System lets peripherals trigger each other directly, bypassing the CPU.
- DMA handles automatic data movement for ADC or comms, freeing the CPU.

1.1.2 Hybrid DSP/MCUs

Devices like TI's C2000 family combine DSP performance with MCU ease-of-use. 32-bit, optimised for high-speed real-time control (motor control, digital power supplies, solar).

1.2 The ARM Architecture

- ARM Holdings licenses its ISA and architecture; manufacturers integrate their own peripherals.
- RISC-based design uses fewer transistors → lower power and heat → dominant in mobile.

1.2.1 Cortex-M Hierarchy

Family	Notes
Cortex-M0+	Lowest power; simple IoT; replaces 8-bit MCUs
Cortex-M3	General embedded control; hardware divide
Cortex-M4	DSP and FPU; motor drives and signal processing
Cortex-M7	High-performance real-time; dual-issue pipeline
Cortex-A	MMU enables full OSs like Linux (e.g., Raspberry Pi)

Choose ARM over a simpler MCU (e.g., AT90USB) when you need FPU math, more than 128 KB Flash or 8 KB RAM, or faster comms (Full-Speed USB, Ethernet).

1.3 MCUs vs. MPUs

Feature	Microcontroller (MCU)	Microprocessor (MPU)
Integration	CPU, memory, I/O on one chip	CPU only; needs external DRAM/NAND/I/O/oscillator
Memory	Up to ~2 MB Flash, 8 KB RAM	External: GBs available
Performance	Real-time OS and simple UIs; deterministic IRQ latency	Full OS, Gigabit Ethernet, GUIs (e.g., 400 DMIPS for Linux)
Real-time	Sub-μs interrupt latency	OS scheduler adds jitter; no hard real-time
Complexity	Single-rail supply; few external components	Multi-rail supplies; high PCB complexity (6+ layers, impedance control)
Power / cost	Sub-μA sleep; \$1–\$10	Higher power; \$10–\$100+

1.4 Selection Advice for Advanced Systems

- Use what you know unless the project demands a new architecture.
- If code doesn't fit, buy a bigger MCU rather than fighting extreme assembly optimisations.
- Hybrid solutions (e.g., TI F28M3X dual-core: ARM Cortex-M3 + C2000) prevent the OS interfering with high-speed control loops.
- MPUs shine when an OS like Linux can supply built-in drivers (TCP/IP, USB, video); bare-metal MPU work is a huge undertaking.

1.5 Use Cases & Applications

1.5.1 Where each device family lives

Device family	Typical product
AT90USB / 8-bit AVR	Battery sensor nodes, simple appliances, hobby USB gadgets
XMEGA (8/16-bit hybrid)	Higher-precision sensor frontends with DMA-driven ADC
Cortex-M0+/M3	IoT modules, fitness trackers, mid-range consumer products
Cortex-M4 + FPU	Motor drives, drones, audio DSP
Cortex-M7	Advanced industrial controllers, image preprocessing
Cortex-A (Linux MPU)	Single-board computers, set-top boxes, 7" smart-home controllers
TI C2000 / DSP-MCU	Solar inverters, power converters, sensorless motor control

1.6 Decision Framework

- If your code fits in <128 KB Flash and 8 KB RAM, real-time deadlines exist, and you don't need TCP/IP → an 8-bit/16-bit MCU is usually cheapest.
- If you need FPU / DSP, USB, Ethernet, or larger memory but still want hard real-time → ARM Cortex-M.
- If you need a full operating system (Linux), gigabit Ethernet, or a rich GUI → MPU; accept added complexity, power, and PCB cost.
- If you need both real-time control AND an OS → hybrid devices like TI F28M3X (M3 + C2000) keep concerns separate.

1.7 Worked Example: Picking a Chip for a Quadcopter Flight Controller

- Sensor fusion (gyro + accel + magnetometer) requires float math and 1–2 kHz update rates → Cortex-M4 with FPU is ideal.
- Logging to SD card and a 100 Mbps Ethernet uplink would push toward Cortex-M7 or a dual-core part.
- If you also want a desktop-style GUI for live telemetry on the device itself, Linux on a Cortex-A is justified.

Key Definitions

MCU
: Single-chip computer integrating CPU, memory (Flash/RAM), and I/O peripherals.

MPU
: CPU-focused chip needing external memory, I/O, and support components.

ARM
: Acorn RISC Machines; licensed ISA dominant in mobile due to low power per watt.

RISC
: Reduced Instruction Set Computer; small, simple instruction set with low transistor count.

DMA
: Direct Memory Access; hardware that moves data between peripherals and memory without CPU.

Event System
: Peripheral-to-peripheral hardware triggering used in advanced MCUs (e.g., XMEGA).

DSP
: Digital Signal Processor optimised for multiply-accumulate / signal processing.

FPU
: Floating-Point Unit; coprocessor for high-speed floating-point arithmetic.

DMIPS
: Dhrystone MIPS; CPU performance benchmark; Linux on ARM typically wants $\geq$400 DMIPS.

MMU
: Memory Management Unit; required for general-purpose OSs (Linux/Windows) to provide virtual memory and process isolation.

Hybrid Core
: Single chip combining a control-oriented core (e.g., Cortex-M3) with a real-time DSP core (e.g., C2000).

Sources

- Lecture 4.pdf